# Datamodel: allStarLite

**General Description**

The allStarLite-APRED_VERS-ASPCAP_VERS.fits file contains an abbreviated version of the final data product and summary information for every observed star in the data release, i.e. a lighter-weight version of the allStar file. Specifically, it removes the tags ALL_VISITS, VISITS, ALL_VISIT_PK, VISIT_PK, FPARAM_CLASS, CHI2_CLASS, FPARAM, FELEM, FPARAM_COV, PARAM, PARAM_COV, ELEMFLAG, FELEM_ERR, APSTAR_ID, TARGET_ID, ASPCAP_ID, FILE, LOCATION_ID, all of which may not be used by many users; many of them contain data in array form that is duplicated in other tags. The advantage is a file that is 3 times smaller than the allStar file!

**Naming Convention**

`allStarLite-APRED_VERS-ASPCAP_VERS.fits`

**Approximate Size**

1 GB

**File Type**

FITS

**Read by Products**

none

**Written by Products**

idlwrap (aspcap_allstar)

**Sections**


- HDU0: Primary Header
- HDU1: Summary Data Table
- HDU2: Data Index
- HDU3: Array Definitions
- Examples


## HDU0: Primary Header

Required Header Keywords

| KEY | Value | Type | Comment |
|---|---|---|---|
| DATE | 2013-03-17 | str | Creation UTC (CCCC-MM-DD) date of FITS header |

## HDU1: Summary data table

Contains an entry for every star in the data release. Relays information about the combined spectrum and derived quantities for each DR star, in particular, radial velocity values, stellar atmospheric parameters, and individual element abundances as determined from ASPCAP analysis.

Required Data Table Columns

| Name | Type | Units | Description |
|---|---|---|---|
| APOGEE_ID | char[18] | | TMASS-STYLE object name |
| TELESCOPE | char[8] | | String representation of of telescope used for observation (currently APO 1m or 2.5m) |
| FIELD | char[16] | | Field name |
| J | float32 | | 2MASS J mag [bad=99] |
| J_ERR | float32 | | Uncertainty in 2MASS J mag |
| H | float32 | | 2MASS H mag [bad=99] |
| H_ERR | float32 | | Uncertainty in 2MASS H mag |
| K | float32 | | 2MASS Ks mag [bad=99] |
| K_ERR | float32 | | Uncertainty in 2MASS Ks mag |
| RA | float64 | degrees | Right ascension (J2000) |
| DEC | float64 | degrees | Declination (J2000) |
| GLON | float64 | degrees | Galactic longitude |
| GLAT | float64 | degrees | Galactic latitude |
| APOGEE_TARGET1 | int32 | | Bitwise OR of first APOGEE-1 target flag of all visits, see bitmask definitions. |

| | | | |
|---|---|---|---|
| APOGEE_TARGET2 | int32 | | Bitwise OR of second APOGEE-1 target flag of all visits, see bitmask definitions |
| APOGEE_TARGET3 | int32 | | Non-populated/dummy third APOGEE-1 targeting flag |
| APOGEE2_TARGET1 | int32 | | Bitwise OR of first APOGEE-2 target flag of all visits, see bitmask definitions. |
| APOGEE2_TARGET2 | int32 | | Bitwise OR of second APOGEE-2 target flag of all visits, see bitmask definitions |
| APOGEE2_TARGET3 | int32 | | Bitwise OR of third APOGEE-2 target flag of all visits, see bitmask definitions |
| TARGFLAGS | char[116] | | Verbose/text form of APOGEE-1 target flags |
| SURVEY | char[35] | | Survey-associated with object: apogee, apo1m, apogee-marvels, apogee2, apogee2-manga, manga-apogee2 |
| PROGRAMNAME | char[18] | | Program name associated with object, when available |
| NINST | int32 | Three-entry array which indicates the number of object visits in commissioning, APOGEE-1, APOGEE-2 | |
| NVISITS | int32 | | Number of visits into combined spectrum |
| COMBTYPE | int32 | | Gives type of RV combination used |
| COMMISS | int16 | | Set to 1 for APOGEE-1 Commissioning data (before July 2011), else 0 |
| SNR | float32 | | median S/N per pixel in combined frame (at apStar sampling) |
| STARFLAG | int32 | | Flag for star condition taken from bitwise OR of individual visits, see bitmask definitions |
| STARFLAGS | char[129] | | Verbose/text form of STARFLAG |
| ANDFLAG | int32 | | Flag for star condition taken from bitwise AND of individual visits, see bitmask definitions |
| ANDFLAGS | char[59] | | Verbose/text form of ANDFLAG |
| VHELIO_AVG | float32 | km/s | Average radial velocity, weighted by S/N, using RVs determined from cross-correlation of individual spectra with combined spectrum |
| VSCATTER | float32 | km/s | Scatter of individual visit RVs around average |
| VERR | float32 | km/s | Uncertainty in VHELIO_AVG from the S/N-weighted individual RVs |
| VERR_MED | float32 | km/s | Median of individual visit RV errors |
| OBSVHELIO_AVG | float32 | km/s | Average RV from observed spectrum template matching technique |
| OBSVSCATTER | float32 | km/s | Scatter of individual visit observed RVs around average |
| OBSVERR | float32 | km/s | S/N weighted, pipeline-reported error in the observed RV's |
| OBSVERR_MED | float32 | km/s | Median of individual visit observed RV errors |
| SYNTHVHELIO_AVG | float32 | km/s | Average RV from synthetic spectrum template matching technique |
| SYNTHVSCATTER | float32 | km/s | Scatter of individual visit synthethic RVs around average |
| SYNTHVERR | float32 | km/s | S/N weighted, pipeline-reported error in the synthetic RV's |
| SYNTHVERR_MED | float32 | km/s | Median of individual visit synthetic RV errors |
| RV_TEFF | float32 | K | Teff of best-match synthetic spectrum from RV grid (NOT ASPCAP!) |

| | | | |
|---|---|---|---|
| RV_LOGG | float32 | log (cgs) | log g of best-match synthetic spectrum from RV grid (NOT ASPCAP!) |
| RV_FEH | float32 | | [Fe/H] of best-match synthetic spectrum from RV grid (NOT ASPCAP!) |
| RV_ALPHA | float32 | | [alpha/M] of best-match synthetic spectrum from RV grid (NOT ASPCAP!) |
| RV_CARB | float32 | | [C/M] of best-match synthetic spectrum from RV grid (NOT ASPCAP!) |
| RV_CCFWHM | float32 | km/s | FWHM of cross-correlation peak from combined vs best-match synthetic spectrum |
| RV_AUTOFWHM | float32 | km/s | FWHM of auto-correlation of best-match synthetic spectrum |
| SYNTHSCATTER | float32 | | scatter between RV using combined spectrum and RV using synthetic spectrum |
| STABLERV_CHI2 | float32[2] | | Chi^2 of RV distribution under assumption of a stable single-valued RV; perhaps not currenlty useful because of issues with understanding RV errors |
| STABLERV_RCHI2 | float32[2] | | Reduced chi^2 of RV distribution |
| CHI2_THRESHOLD | float32[2] | | Threshold chi^2 for possible binary determination (not currently valid) |
| STABLERV_CHI2_PROB | float32[2] | | Probability of obtaining observed chi^2 under assumption of stable RV |
| MEANFIB | float32 | | Mean fiber number of the set of observations |
| SIGFIB | float32 | | Dispersion in fiber number |
| SNREV | float32 | | Revised S/N estimate (avoiding persistence issues) |
| APSTAR_VERSION | char[5] | | APSTAR release version |
| ASPCAP_VERSION | char[10] | | ASPCAP release version |
| RESULTS_VERSION | char[8] | | RESULTS release version |
| EXTRATARG | int32 | | Bitmask which identifies main survey targets and other classes, see bitmask definitions. |
| MIN_H | | | Bright H limit for target selection for this object |
| MAX_H | | | Faint H limit for target selection for this object |
| MIN_JK | | | Blue (J-K) limit for target selection for this object |
| MAX_JK | | | Red (J-K) limit for target selection for this object |
| TEFF | float32 | K | Teff from ASPCAP analysis of combined spectrum (from PARAM) |
| TEFF_ERR | float32 | K | Teff uncertainty (from PARAM_COV) |
| LOGG | float32 | log (cgs) | log g from ASPCAP analysis of combined spectrum (from PARAM) |
| LOGG_ERR | float32 | log (cgs) | log g uncertainty (from PARAM_COV) |
| VMICRO | float32 | (cgs) | microturbulent velocity (fit for dwarfs, f(log g) for giants) |
| VMACRO | float32 | (cgs) | macroturbulent velocity (f(log Teff, [M/H]) for giants) |
| VSINI | float32 | (cgs) | rotational+macroturbulent velocity (fit for dwarfs) |
| M_H | float32 | dex | [Z/H] from ASPCAP analysis of combined spectrum (from PARAM) |
| M_H_ERR | float32 | dex | [Z/H] uncertainty (from PARAM_COV) |
| ALPHA_M | float32 | dex | [alpha/M] from ASPCAP analysis of combined spectrum (from PARAM) |
| ALPHA_M_ERR | float32 | dex | [alpha/M] uncertainty (from PARAM_COV) |
| ASPCAP_CHI2 | float32 | | Chi^2 from ASPCAP fit |
| ASPCAP_CLASS | char[2] | | Temperature class of best-fitting spectrum |
| ASPCAPFLAG | int32 | | Flag for ASPCAP analysis, see bitmask definitions |

| | | | |
|---|---|---|---|
| ASPCAPFLAGS | char[114] | | Verbose/text form ASPCAPFLAG |
| PARAMFLAG | int32[7] | | Individual parameter flag for ASPCAP analysis, see bitmask definitions |
| X_H | float32[26] | | Empirically calibrated individual element array, using ASPCAP stellar abundances fit + calibrations, all expressed in logarithmic abundance relative to H ([X/H]), in order given in ELEM_SYMBOL array in HDU3 |
| X_H_ERR | float32[26] | | Empirical uncertainties in [X/H], derived from repeat observations of stars |
| X_M | float32[26] | | Empirically calibrated individual element array, using ASPCAP stellar abundances fit + calibrations, all expressed in logarithmic abundance relative to M ([X/M]) in order given in ELEM_SYMBOL array in HDU3 |
| X_M_ERR | float32[26] | | Empirical uncertainties in [X/M], derived from repeat observations of stars |
| C_FE | float32 | dex | [C/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| CI_FE | float32 | dex | [CI/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| N_FE | float32 | dex | [N/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| O_FE | float32 | dex | [O/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| NA_FE | float32 | dex | [Na/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| MG_FE | float32 | dex | [Mg/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| AL_FE | float32 | dex | [Al/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| SI_FE | float32 | dex | [Si/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| P_FE | float32 | dex | [P/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| S_FE | float32 | dex | [S/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| K_FE | float32 | dex | [K/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| CA_FE | float32 | dex | [Ca/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| TI_FE | float32 | dex | [Ti/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| TIII_FE | float32 | dex | [TiII/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| V_FE | float32 | dex | [V/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| CR_FE | float32 | dex | [Cr/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| MN_FE | float32 | dex | [Mn/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| FE_H | float32 | dex | [Fe/H] from ASPCAP analysis of combined spectrum (from X_M) |
| CO_FE | float32 | dex | [Co/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| NI_FE | float32 | dex | [Ni/Fe] from ASPCAP analysis of combined spectrum (from X_M) |

| | | | |
|---|---|---|---|
| CU_FE | float32 | dex | [Cu/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| GE_FE | float32 | dex | [Ge/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| RB_FE | float32 | dex | [Rb/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| YB_FE | float32 | dex | [Y/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| ND_FE | float32 | dex | [Nd/Fe] from ASPCAP analysis of combined spectrum (from X_M) |
| C_FE_ERR | float32 | dex | [C/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| CI_FE_ERR | float32 | dex | [CI/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| N_FE_ERR | float32 | dex | [N/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| O_FE_ERR | float32 | dex | [O/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| NA_FE_ERR | float32 | dex | [Na/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| MG_FE_ERR | float32 | dex | [Mg/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| AL_FE_ERR | float32 | dex | [Al/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| SI_FE_ERR | float32 | dex | [Si/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| P_FE_ERR | float32 | dex | [P/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| S_FE_ERR | float32 | dex | [S/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| K_FE_ERR | float32 | dex | [K/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| CA_FE_ERR | float32 | dex | [Ca/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| TI_FE_ERR | float32 | dex | [Ti/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| TIII_FE_ERR | float32 | dex | [TiII/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| V_FE_ERR | float32 | dex | [V/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| CR_FE_ERR | float32 | dex | [Cr/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| MN_FE_ERR | float32 | dex | [Mn/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| FE_H_ERR | float32 | dex | [Fe/H] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| CO_FE_ERR | float32 | dex | [Co/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| NI_FE_ERR | float32 | dex | [Ni/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| CU_FE_ERR | float32 | dex | [Cu/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| GE_FE_ERR | float32 | dex | [Ge/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| RB_FE_ERR | float32 | dex | [Rb/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |

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|---|---|---|---|
| YB_FE_ERR | float32 | dex | [Y/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| ND_FE_ERR | float32 | dex | [Nd/Fe] uncertainty from ASPCAP analysis of combined spectrum (from X_M) |
| C_FE_FLAG | float32 | dex | [C/Fe] flag |
| CI_FE_FLAG | float32 | dex | [CI/Fe] flag |
| N_FE_FLAG | float32 | dex | [N/Fe] flag |
| O_FE_FLAG | float32 | dex | [O/Fe] flag |
| NA_FE_FLAG | float32 | dex | [Na/Fe] flag |
| MG_FE_FLAG | float32 | dex | [Mg/Fe] flag |
| AL_FE_FLAG | float32 | dex | [Al/Fe] flag |
| SI_FE_FLAG | float32 | dex | [Si/Fe] flag |
| P_FE_FLAG | float32 | dex | [P/Fe] flag |
| S_FE_FLAG | float32 | dex | [S/Fe] flag |
| K_FE_FLAG | float32 | dex | [K/Fe] flag |
| CA_FE_FLAG | float32 | dex | [Ca/Fe] flag |
| TI_FE_FLAG | float32 | dex | [Ti/Fe] flag |
| TIII_FE_FLAG | float32 | dex | [TiII/Fe] flag |
| V_FE_FLAG | float32 | dex | [V/Fe] flag |
| CR_FE_FLAG | float32 | dex | [Cr/Fe] flag |
| MN_FE_FLAG | float32 | dex | [Mn/Fe] flag |
| FE_H_FLAG | float32 | dex | [Fe/H] flag |
| CO_FE_FLAG | float32 | dex | [Co/Fe] flag |
| NI_FE_FLAG | float32 | dex | [Ni/Fe] flag |
| CU_FE_FLAG | float32 | dex | [Cu/Fe] flag |
| GE_FE_FLAG | float32 | dex | [Ge/Fe] flag |
| RB_FE_FLAG | float32 | dex | [Rb/Fe] flag |
| YB_FE_FLAG | float32 | dex | [Y/Fe] flag |
| ND_FE_FLAG | float32 | dex | [Nd/Fe] flag |
| ELEM_CHI2 | float32[26] | | Chi^2 from ASPCAP fit of individual abundances |
| ALT_ID | char[18] | | Quasi-unique identifier: (HEY: The 1m targets have a blank string) |
| SRC_H | char[12] | | Source of H-Band photometry |
| WASH_M | float32 | | Washington M mag |
| WASH_M_ERR | float32 | | Washington M mag error |
| WASH_T2 | float32 | | Washington T2 mag |
| WASH_T2_ERR | float32 | | Washington T2 mag error |
| DDO51 | float32 | | DDO 51 mag |
| DDO51_ERR | float32 | | DDO 51 mag error |
| IRAC_3_6 | float32 | | IRAC 3.6micron mag |
| IRAC_3_6_ERR | float32 | | IRAC 3.6micron mag error |
| IRAC_4_5 | float32 | | IRAC 4.5micron mag |
| IRAC_4_5_ERR | float32 | | IRAC 4.5micron mag error |
| IRAC_5_8 | float32 | | IRAC 5.8 micron mag |
| IRAC_5_8_ERR | float32 | | IRAC 5.8 micron mag error |
| IRAC_8_0 | float32 | | IRAC 8.0 micron mag |
| IRAC_8_0_ERR | float32 | | IRAC 8.0 micron mag error |

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|---|---|---|---|
| WISE_4_5 | float32 | | WISE 4.5 micron mag |
| WISE_4_5_ERR | float32 | | WISE 4.5 micron mag error |
| TARG_4_5 | float32 | | 4.5 micron mag adopted for dereddening for targeting |
| TARG_4_5_ERR | float32 | | 4.5 micron mag adopted for dereddening for targeting, error |
| AK_TARG | float32 | | K-band extinction adopted for targetting |
| AK_TARG_METHOD | char[17] | | Method used to get targetting extinction |
| AK_WISE | float32 | | WISE all-sky K-band extinction |
| SFD_EBV | float32 | | SFD reddening |
| WASH_DDO51_GIANT_FLAG | int16 | | Flagged as a giant for targeting purposes based on Washington/DDO 51 photometry |
| WASH_DDO51_STAR_FLAG | int16 | | Flagged as a starfor targeting purposes based on Washington/DDO 51 photometry |
| GAIA_SOURCE_ID | int64 | | GAIA source ID from GAIA DR2 |
| GAIA_PARALLAX | float64 | mas | GAIA parallax from GAIA DR2 |
| GAIA_PARALLAX_ERROR | float64 | mas | GAIA parallax uncertainty GAIA DR2 |
| GAIA_PMRA | float64 | mas/yr | GAIA proper motion in RA from GAIA DR2 |
| GAIA_PMRA_ERROR | float64 | mas/yr | GAIA uncertainty in proper motion in RA from GAIA DR2 |
| GAIA_PMDEC | float64 | mas/yr | GAIA proper motion in DEC from GAIA DR2 |
| GAIA_PMDEC_ERROR | float64 | mas/yr | GAIA uncdertainty in proper motion in DEC from GAIA DR2 |
| GAIA_PHOT_G_MEAN_MAG | float32 | | GAIA g mag from GAIA DR2 |
| GAIA_PHOT_BP_MEAN_MAG | float32 | | GAIA Bp mag from GAIA DR2 |
| GAIA_PHOT_RP_MEAN_MAG | float32 | | GAIA Rp mag from GAIA DR2 |
| GAIA_RADIAL_VELOCITY | float64 | km/s | GAIA radial velocity from GAIA DR2 |
| GAIA_RADIAL_VELOCITY_ERROR | float64 | km/s | GAIA uncertainty in radial velocity from GAIA DR2 |
| GAIA_R_EST | float64 | pc | GAIA Bailer-Jones distance estimate r_est from GAIA DR2 |
| GAIA_R_LO | float64 | pc | GAIA Bailer-Jones 16th percentile distance r_lo from GAIA DR2 |
| GAIA_R_HI | float64 | pc | GAIA Bailer-Jones 84th percentile distance r_hi from GAIA DR2 |
| TEFF_SPEC | float32 | K | ASPCAP spectroscopic Teff (duplicated from FPARAM[0] for convenience) |
| LOGG_SPEC | float32 | log (cgs) | ASPCAP spectroscopic surface gravity (duplicated from FPARAM[1] for convenience) |

## HDU2: Index to summary data table

Provides RA-indexed array giving index in HDU1 corresponding to each degree in RA, e.g., ARRAY[90] gives the index in HDU1 of first object with RA>90 degrees.

This HDU has no non-standard required keywords.

Data: FITS image

## HDU3: array definition

Provides detailed information about the contents of the PARAM and ELEM arrays in HDU1 that is common to all data release stars.

This HDU has no non-standard required keywords.

Data: FITS image

Required Data Table Columns

| Name | Type | Units | Description |
|---|---|---|---|

| | | |
|---|---|---|
| PARAM_SYMBOL | char[] | Gives order of parameters for PARAM and FPARAM arrays in HDU1 |
| ELEM_SYMBOL | char[] | Gives order of elements for X_H, X_M and FELEM arrays in HDU1 |
| ELEM_VALUE | char[] | Gives nature of values in FELEM arrays in HDU1, i.e. [X/H] or [X/M] |
| ELEMTOH | int[] | Set to 1 for FELEM array element that is relative to H, 0 if relative to M (note caveat for C and N, which are relative to H in the dwarf grids) |
| CLASSES | char[] | Gives order of classes in FPARAM_CLASS arrays in HDU1 |